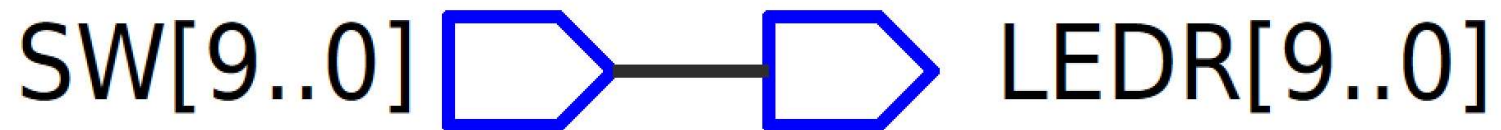
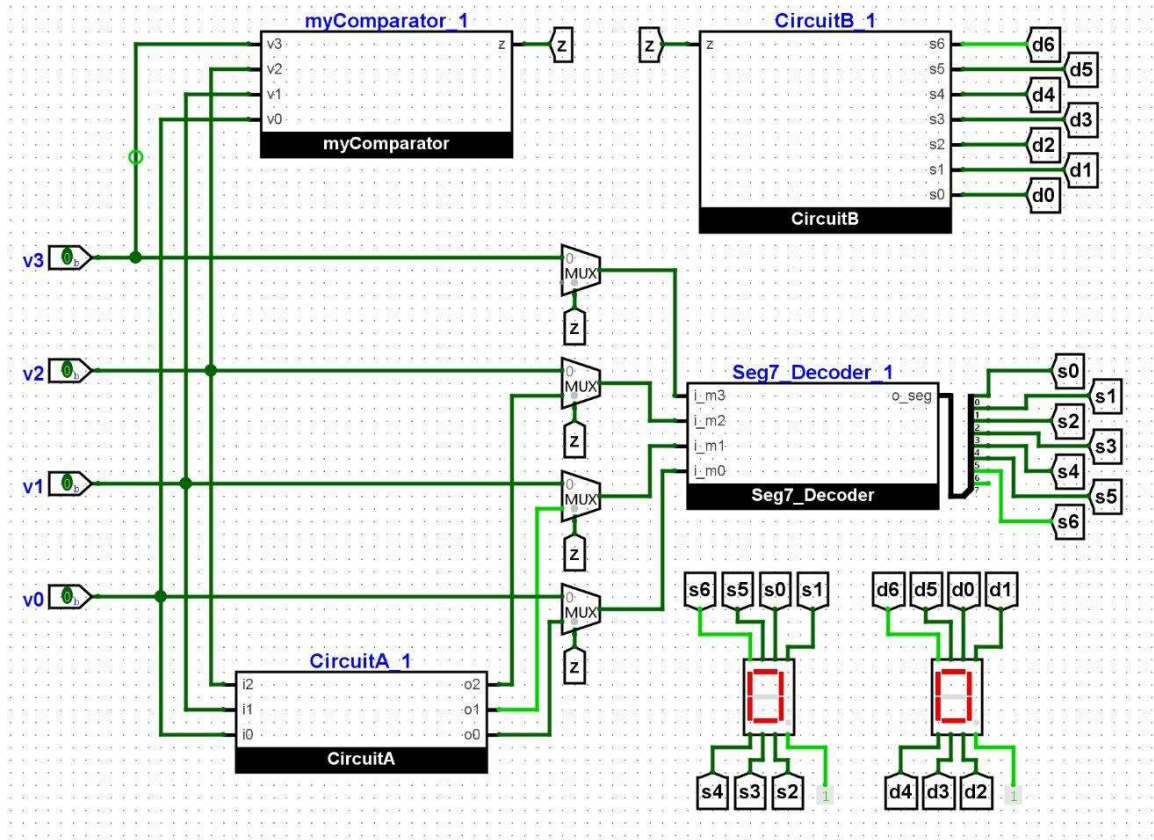


Part I

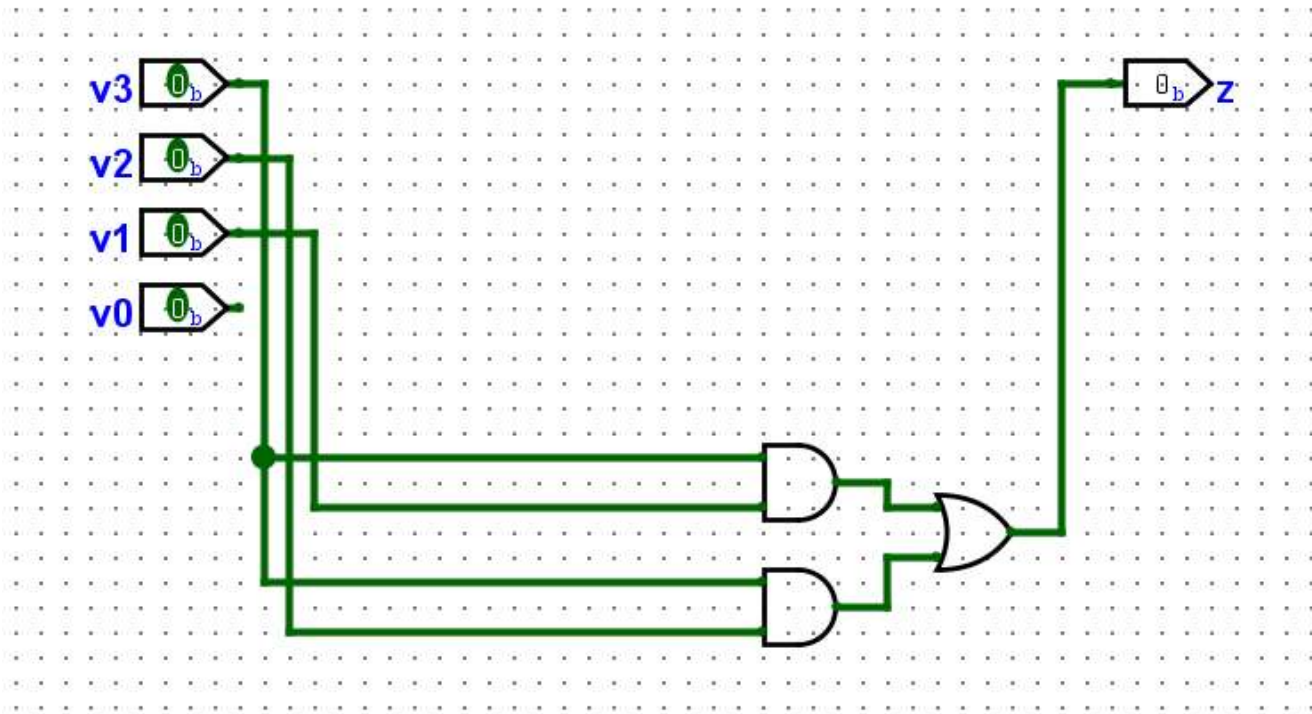
```
]module main(  
    // Pinout assignment  
    input [9:0] SW,  
    output [9:0] LEDR  
);  
  
    // Part I  
    assign LEDR[9:0] = SW[9:0];
```



Part II

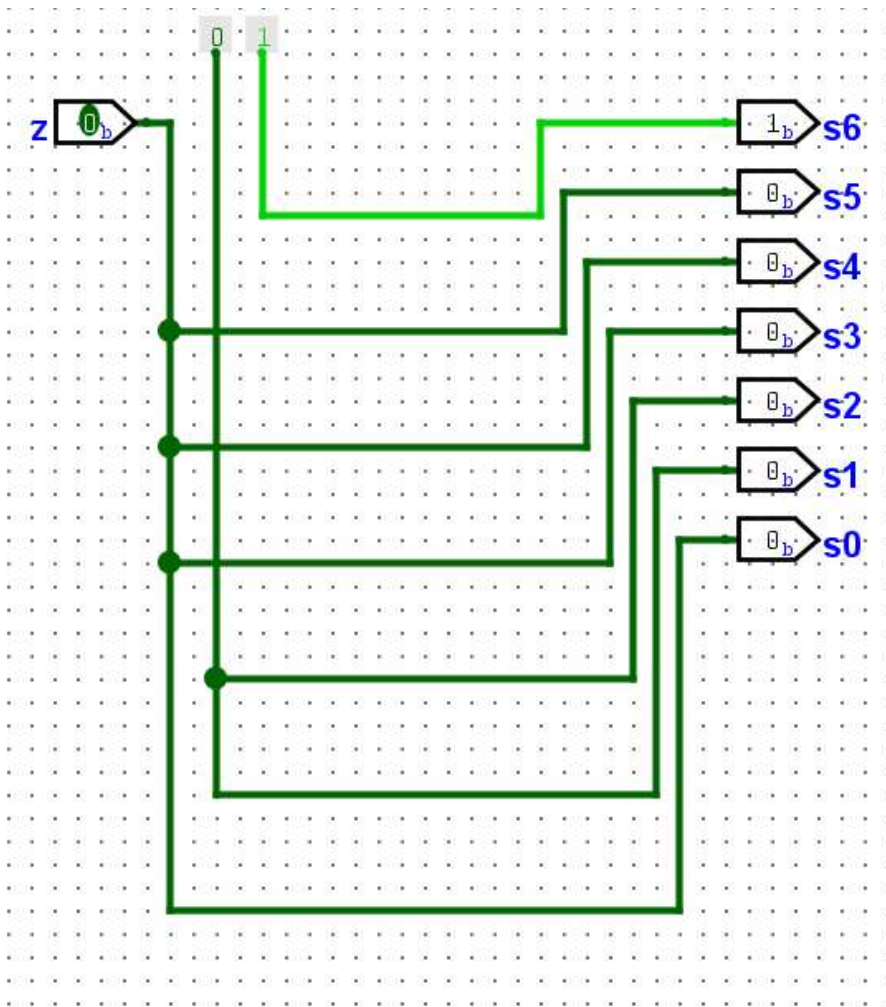


Comparator & Truth Table



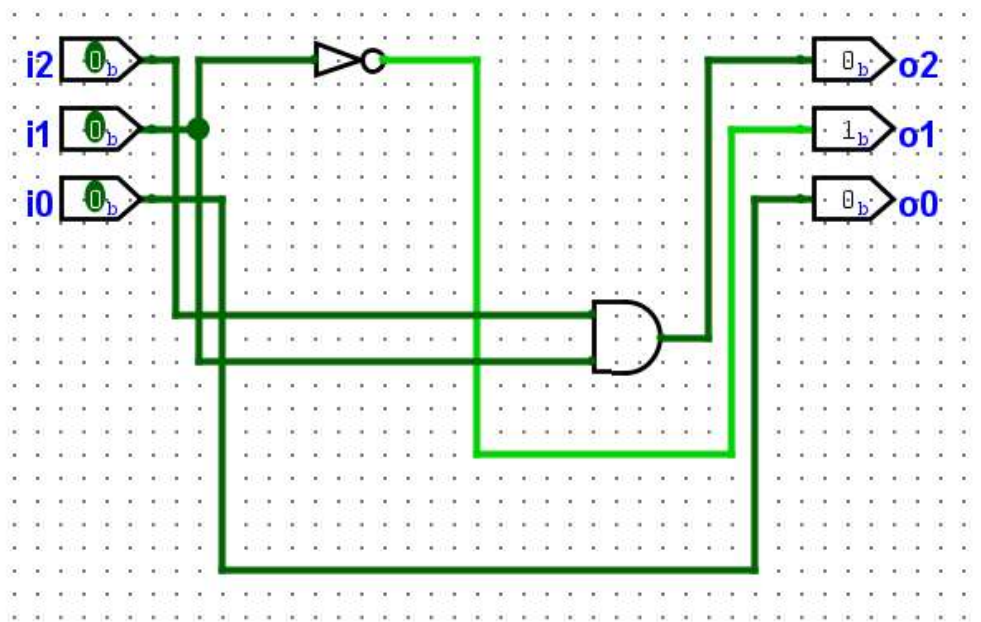
v3	v2	v1	v0	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Circuit B & Truth Table



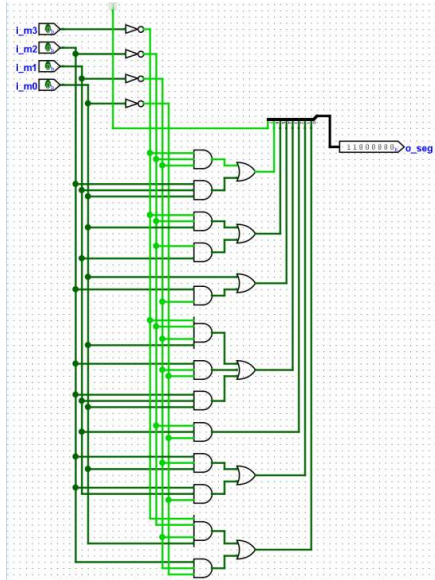
z	s6	s5	s4	s3	s2	s1	s0
0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	1

Circuit A & Truth Table



i_2	i_1	i_0	o_2	o_1	o_0
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	0	1

Segment 7 Decoder

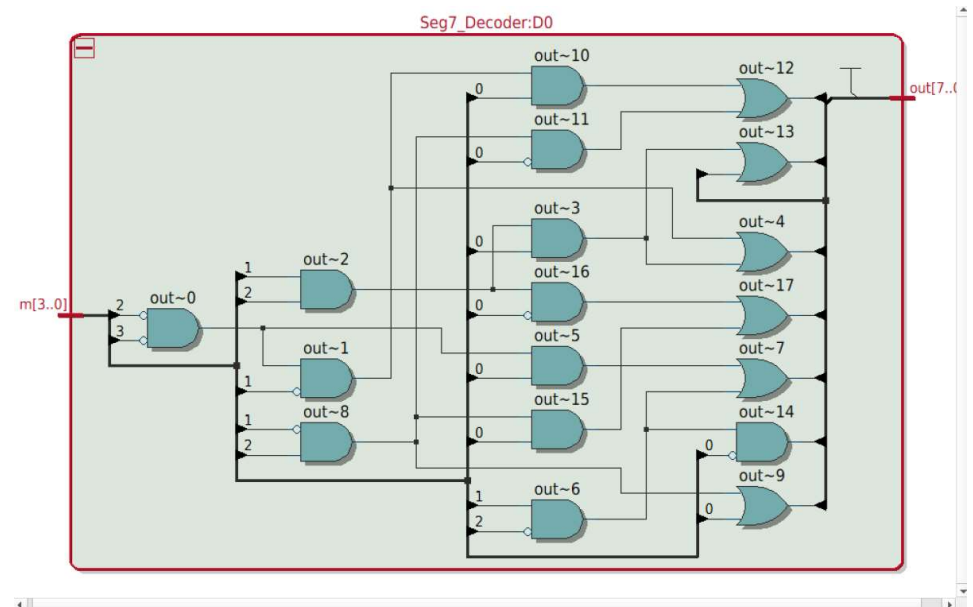


i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 1 1 1 0 0 1
0	0	1	0	1 0 1 0 0 1 0 0
0	0	1	1	1 0 1 1 0 0 0 0
0	1	0	0	1 0 0 1 1 0 0 1
0	1	0	1	1 0 0 1 0 0 1 0
0	1	1	0	1 0 0 0 0 0 1 0
0	1	1	1	1 1 0 1 1 0 0 0
1	0	0	0	1 0 0 0 0 0 0 0
1	0	0	1	1 0 0 1 0 0 0 0
1	0	1	0	1 0 1 0 0 1 0 0
1	0	1	1	1 0 1 1 0 0 0 0
1	1	0	0	1 0 0 1 1 0 0 1
1	1	0	1	1 0 0 1 0 0 1 0
1	1	1	0	1 0 0 0 0 0 1 0
1	1	1	1	1 1 0 1 1 0 0 0

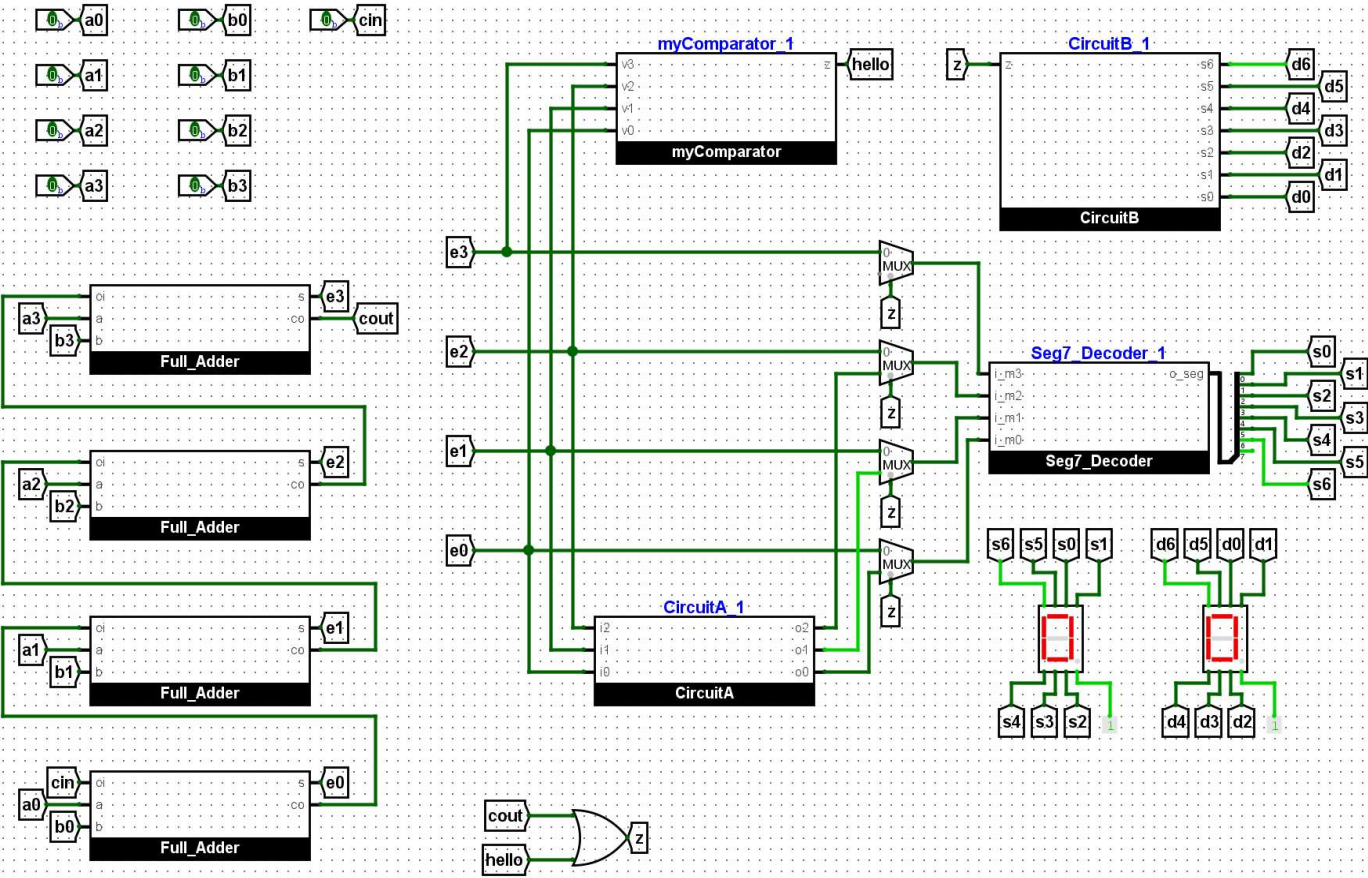
```

1 module Seg7_decoder(
2     input [3:0] m,
3     output [7:0] out
4 );
5
6     //
7     assign out [7] = 1;
8     assign out [6] = ~m[3]&~m[2]&~m[1] | m[2]&m[1]&m[0];
9     assign out [5] = ~m[3]&~m[2]&m[0] | ~m[2]&m[1];
10    assign out [4] = m[0] | m[2]&~m[1];
11    assign out [3] = ~m[3]&~m[2]&~m[1]&m[0] | m[2]&~m[1]&~m[0] | m[2]&m[1]&m[0];
12    assign out [2] = ~m[2]&m[1]&~m[0];
13    assign out [1] = m[2]&~m[1]&m[0] | m[2]&m[1]&~m[0];
14    assign out [0] = ~m[3]&~m[2]&m[1]&m[0] | m[2]&~m[1]&~m[0];
15
16 endmodule

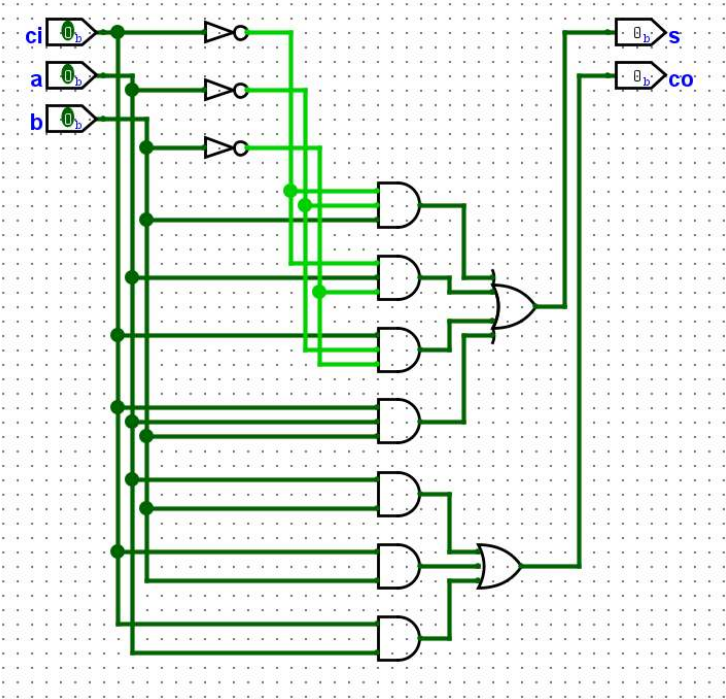
```



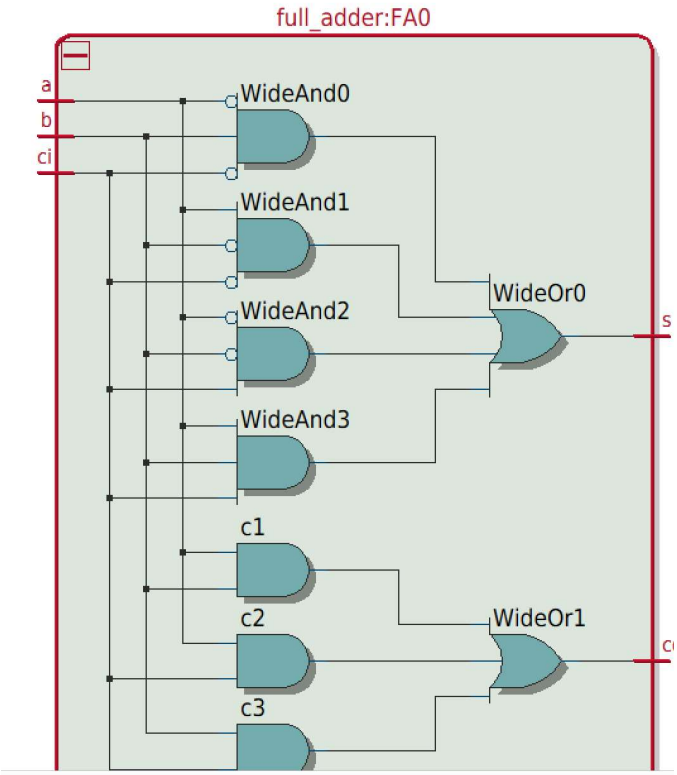
Part III



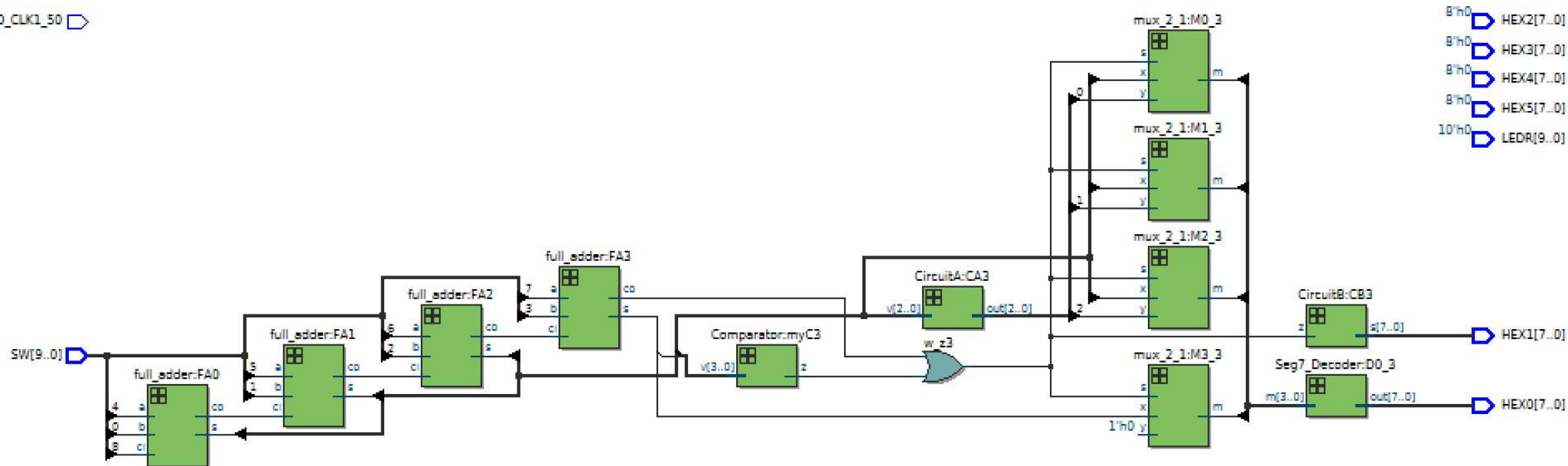
Full Adder Circuit & Truth Table



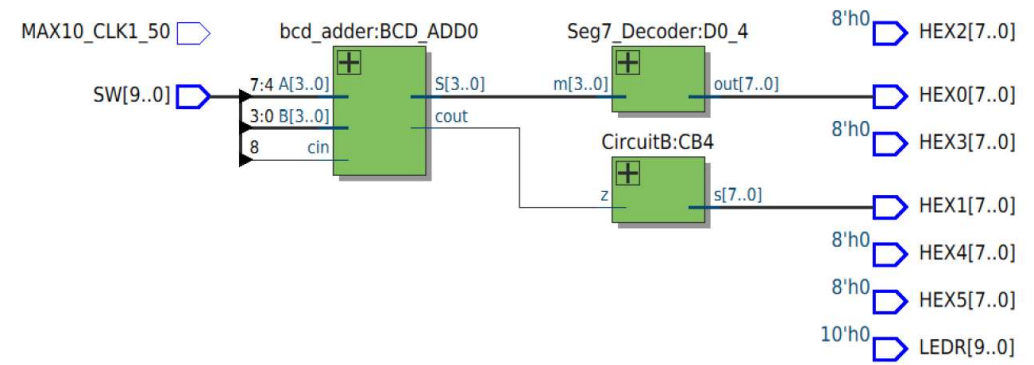
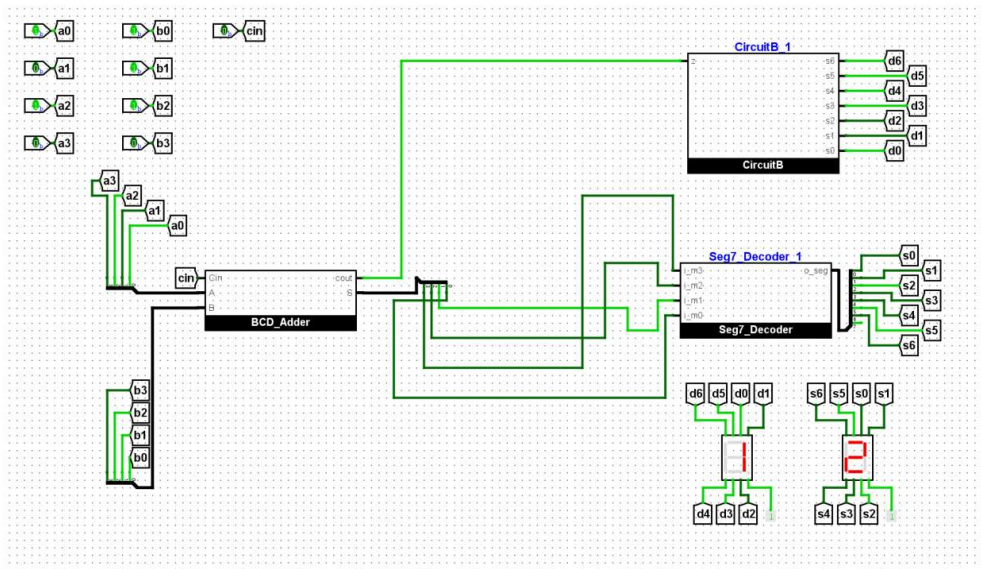
ci	a	b	s	co
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



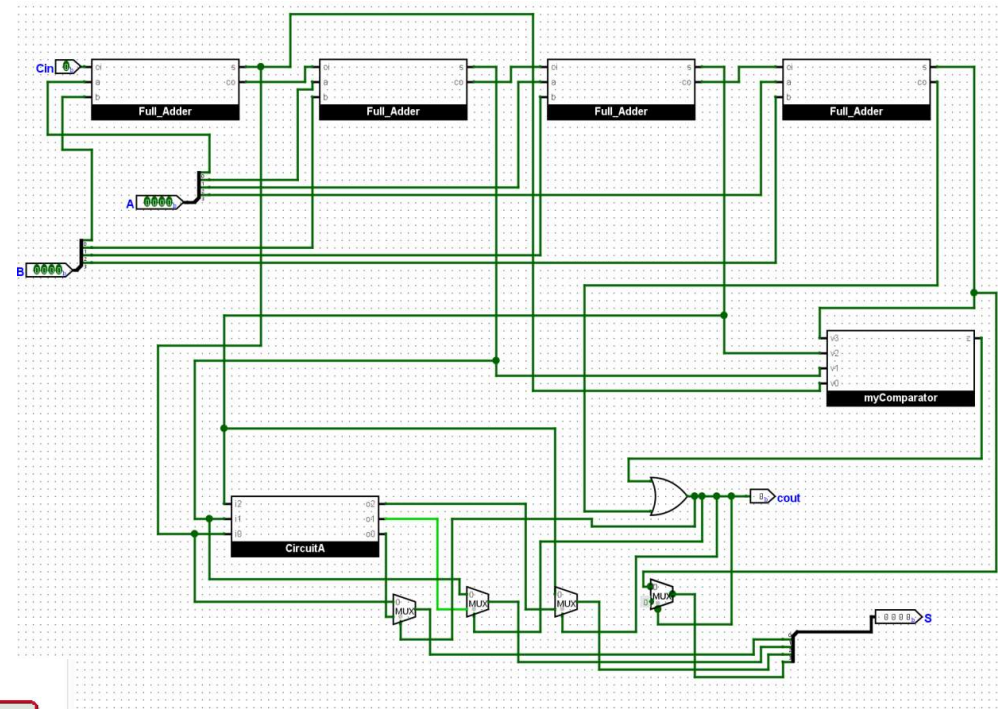
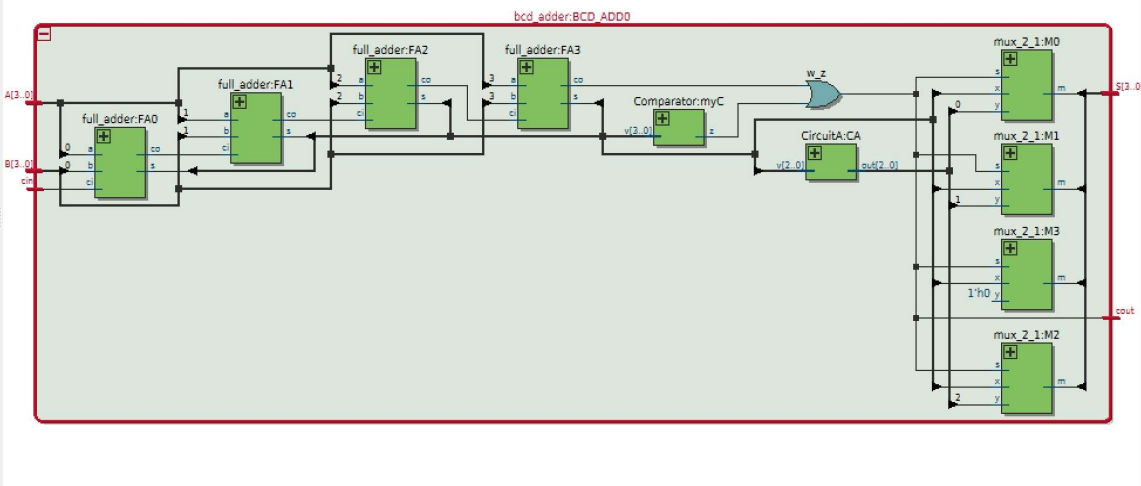
MAX10_CLK1_50



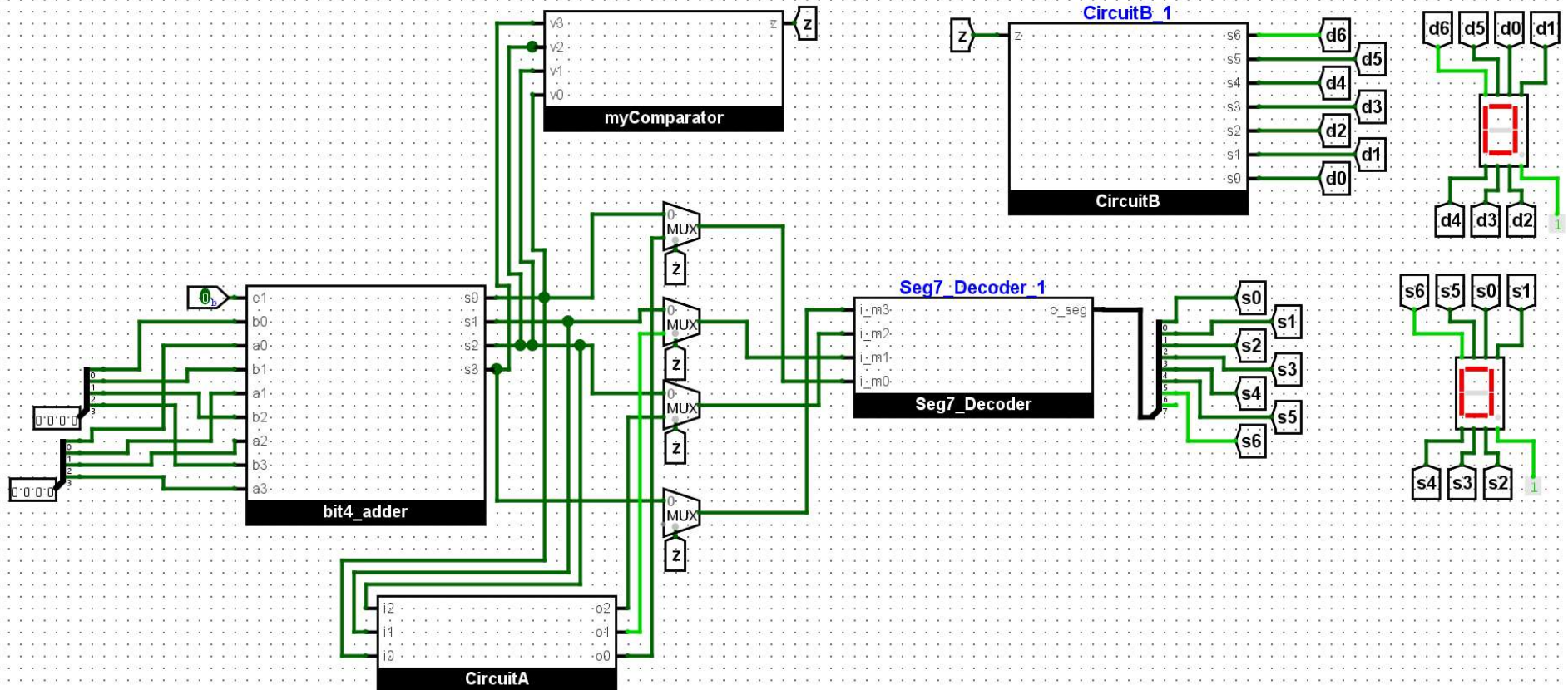
Part IV



BCD Adder



Part V



Bit 4 Adder

